

Network Traffic Analysis of an HTTP Capture

Practice case report | Wireshark | Case002_HTTPCapture

1. Case Details

Examiner	Lucky Bhati
Analysis date	03 October 2026
Case name	Case002_HTTPCapture
Purpose	Practice analysis of a public training capture (no real person's data)

2. Objective

To analyse a packet capture (pcap) of a web browsing session: identify the hosts involved, the protocols used, the DNS lookup, the HTTP requests and responses, and any unusual network behaviour.

3. Tools Used

- Wireshark (packet list, Capture File Properties, Protocol Hierarchy, Conversations, display filters)
- Windows PowerShell (Get-FileHash) for evidence hashing

4. Evidence Details

Evidence file	http.cap (pcap format, Ethernet, 25 kB)
Source	Wireshark Sample Captures (public training file)
Location used	C:\Forensic\Evidence\http.cap
MD5	3346524bd119b437166186b83935ad67
SHA-256	25a72bdf10339f2c29916920c8b9501d294923108de8f29b19aba7cc001ab60d
SHA-1 (shown by Wireshark)	3aac91181c3b7eb34fb7d2b6dd6783f4827fcf07
Hash check	The SHA-256 shown in Wireshark Capture File Properties matches the calculated SHA-256 (Figure 1)

5. Methodology

- Calculated the file hash before analysis.
- Opened http.cap in Wireshark and reviewed all 43 packets.
- Checked Statistics: Capture File Properties, Protocol Hierarchy and Conversations.
- Applied display filters **dns**, **http** and **tcp.analysis.retransmission** to isolate the name lookup, web traffic and abnormal packets.
- Reviewed TCP flags and Wireshark's analysis flags for abnormal packets.
- Recorded findings with packet numbers and screenshots.

6. Findings

6.1 Capture overview. The capture holds 43 packets (25,091 bytes) over 30.394 seconds, recorded on 13 May 2004. Wireshark showed the first and last packets as 15:47:07 and 15:47:37 in the examiner's local time (IST), which is 10:17:07 to 10:17:37 UTC. Of the 43 packets, 41 are TCP and 2 are UDP (DNS). Four packets are HTTP (Figures 1 and 2).

6.2 Hosts and conversations (Figure 3). Three IPv4 conversations were found, all involving the client 145.254.160.237:

Peer	Role	Packets	Bytes	Duration
65.208.228.223	Web server (HTTP, port 80); www.ethereal.com	34	21 kB	30.39 s
145.253.2.203	DNS server (port 53)	2	277 bytes	0.36 s
216.239.59.99	Google ad server (HTTP, port 80)	7	4 kB	1.79 s

6.3 DNS lookup (Figure 4). In packet 13 the client asked the DNS server for the A record of *pagead2.googlesyndication.com*. In packet 17 the server replied with a chain of CNAME records, ending with address records for 216.239.59.104 and 216.239.59.99. The client then contacted 216.239.59.99.

6.4 HTTP activity (Figure 5):

Packet	Direction	Event
1 to 3	Client to 65.208.228.223	TCP three-way handshake (SYN, SYN-ACK, ACK) on port 80
4	Client to 65.208.228.223	GET /download.html, Host: www.ethereal.com; User-Agent shows Mozilla/Firefox on Windows XP
13, 17	Client and DNS server	DNS query and response for pagead2.googlesyndication.com
18	Client to 216.239.59.99	GET /pagead/ads?... (Google ad request)
27	216.239.59.99 to client	HTTP 200 OK (text/html)
38	65.208.228.223 to client	HTTP 200 OK; page content delivered over many TCP segments and reassembled in this packet (about 18 kB of XML/HTML)
40 to 43	Both	TCP connection closed with FIN and ACK packets

6.5 Unusual packets (Figures 6 and 7). Packet 36 is flagged by Wireshark as *TCP Spurious Retransmission*: the ad server (216.239.59.99) sent again a segment that the client had already acknowledged. Packet 37 is flagged as a *TCP Dup ACK*. The display filter **tcp.analysis.retransmission** returned only this one packet (1 of 43 packets, 2.3%). This is normal network behaviour and not an attack indicator. Also, no handshake packets (SYN) for the client port 3371 connection appear in the capture, so that connection was probably already open when the capture began.

7. Conclusion

The capture shows a single browser session on 13 May 2004. The client downloaded the page /download.html from www.ethereal.com and, as part of loading it, looked up and requested a Google advertisement. All requests returned HTTP 200 OK, and no signs of malicious activity were observed. One spurious retransmission and its duplicate acknowledgement were identified.

8. Limitations

- Only the packet list, statistics and filters were reviewed. The Follow TCP Stream view was not used for this report.
- The gzip-compressed ad response was not decompressed or inspected.
- Wireshark shows times in the examiner's local time zone, so the UTC values above were calculated from that.
- The reason for the spurious retransmission was not investigated.
- This is a public training capture, so the findings do not describe a real incident.

9. Chain of Custody Note

Date	Action	By
03 Oct 2026	Obtained http.cap (public sample), calculated hashes, stored in C:\Forensic\Evidence	Lucky Bhati
03 Oct 2026	Opened in Wireshark (analysis only, original file not modified), took screenshots	Lucky Bhati

10. Evidence Screenshots

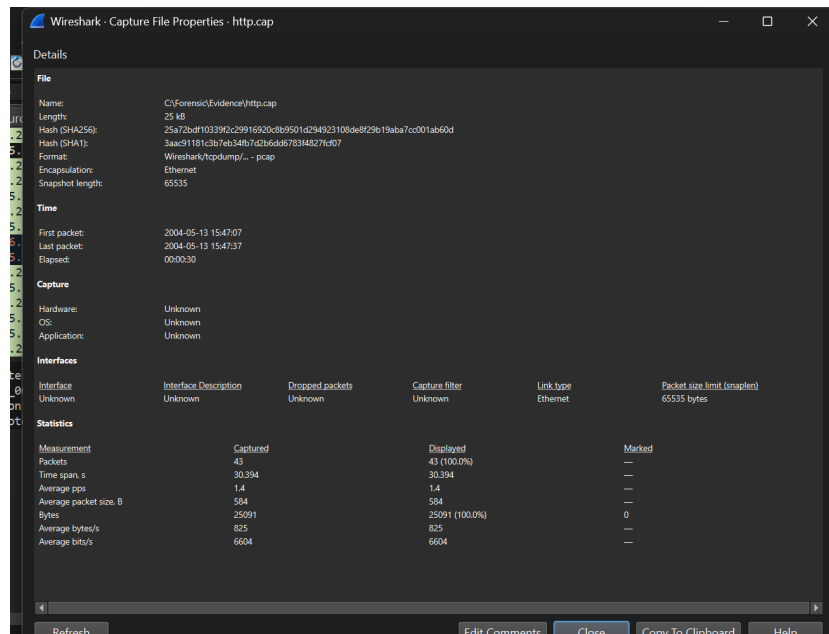


Figure 1: Capture File Properties (hash, time, packet counts)

The screenshot shows the 'Wireshark - Protocol Hierarchy Statistics - http.cap' window. The table displays the following data:

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU
Frame	100.0	43	100.0	25091	6604	0	0	0	43
Ethernet	100.0	43	2.4	602	158	0	0	0	43
Internet Protocol Version 4	100.0	43	3.4	860	226	0	0	0	43
User Datagram Protocol	4.7	2	0.1	16	4	0	0	0	2
Domain Name System	4.7	2	0.8	193	50	2	193	50	2
Transmission Control Protocol	95.3	41	3.3	836	220	37	756	198	41
Hypertext Transfer Protocol	9.3	4	7.2	1812	476	2	1200	315	4
Line-based text data	2.3	1	14.4	3608	949	1	3608	949	1
eXtensible Markup Language	2.3	1	72.0	18070	4756	1	18070	4756	1

Figure 2: Protocol Hierarchy: TCP 41 packets, UDP/DNS 2 packets, HTTP 4 packets

Wireshark - Conversations - http.cap

Conversations Settings

- ☐ Name resolution
- ☒ Absolute start time
- ☒ Display raw data
- ☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

- Bluetooth
- BPV7
- DCCP
- DNP 3.0
- ☒ Ethernet
- FC

Filter list for specific type

Ethernet - 1	IPv4 - 3	IPv6	TCP - 2	UDP - 1
Address A	Address B	Packets	Bytes	Stream ID
145.254.160.237	65.208.228.223	34	21 kB	0
145.254.160.237	145.253.2.203	2	277 bytes	1
145.254.160.237	216.239.59.99	7	4 kB	2

Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits
16	1 kB	18	19 kB	0.000000	30.3937	30.3937
1	89 bytes	1	188 bytes	2.553672	0.3605	197.12
3	883 bytes	4	3 kB	2.984291	1.7926	390.12

Close Help

The image shows a Wireshark packet capture of a DNS transaction. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions like opening files, saving, and analyzing. The packet list pane on the left shows two packets: a Standard query (packet 13) and a Standard query response (packet 17). The packet details pane on the right shows the structure of the selected packet (packet 17), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System. The packet bytes pane at the bottom shows the raw data of the selected packet, with a hex-to-ASCII conversion on the right.

No.	Time	Source	Destination	Protocol	Length	Info
13	2.553672	145.254.160.237	145.253.2.203	DNS	89	Standard query 0x0023 A pagead2.googleusercontent.com
17	2.914190	145.253.2.203	145.254.160.237	DNS	188	Standard query response 0x0023 A pagead2.googleusercontent.com CNAME pagead2.google.com CNAME pagead2.google.com

Frame 17: Packet, 188 bytes on wire (1504 bits), 188 bytes captured (1504 bits) on interface 0
 Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00), Dst: Xerox_00:00:00:00:00:00 (00:00:00:00:00:00)
 Internet Protocol Version 4, Src: 145.253.2.203, Dst: 145.254.160.237
 User Datagram Protocol, Src Port: 53, Dst Port: 3009
 Domain Name System (response)

0000 00 00 01 00 00 00 fe ff 20 00 01 00 00 00 45 00E
 0010 00 ae 15 95 40 00 f9 11 a3 f5 91 fd 02 cb 91 fe@.#.
 0020 00 ad 00 35 00 c1 00 9a 00 02 00 23 81 00 00 015.....
 0030 00 04 00 00 00 00 07 70 61 67 65 61 64 32 11 67p agead2 g
 0040 6f 6f 67 6c 65 73 79 6e 64 69 63 61 74 69 6f 6e ooglesyn dication
 0050 03 63 6f 64 00 00 01 00 01 c0 0c 00 05 00 01 00com
 0060 00 bc c1 00 11 07 70 61 67 65 61 64 32 06 67 6fpa gead2 go
 0070 6f 67 6c 65 c0 26 c0 3b 00 05 00 01 00 00 00 7a ogle &z
 0080 00 1a 00 70 61 67 65 61 64 06 67 6f 67 6c 65pagea d google
 0090 06 61 6b 61 64 6e 73 03 6e 65 74 00 c5 08 00 01 akadns net X
 00a0 00 01 00 00 7b 00 04 d8 ef 3b 68 c0 58 00 01{.....h X
 00b0 00 01 00 00 7b 00 04 d8 ef 3b 63{.....c

Domain Name System: Protocol

Packets: 43 · Displayed: 2 (4.7%)

Profile: Default

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions like opening files, saving, and filtering. The status bar at the bottom indicates 'Packets: 43 · Displayed: 4 (9.3%)' and 'Profile: Default'.

The packet list pane shows a single packet selected:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.911510	145.254.160.237	65.208.228.223	HTTP	533	GET /download.html HTTP/1.1

The packet details pane for the selected packet shows the following structure:

- Ethernet II, Src: Xerox_00:00:00:00:00:00, Dst: fe:ff:20:00:01:00 (fe:ff:20:00:01:00)
- Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.208.228.223
- Transmission Control Protocol, Src Port: 3372, Dst Port: 80, Seq: 1, Ack: 1, Len: 479
- Hypertext Transfer Protocol

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII column displays the following text:

```

GET /download.html HTTP/1.1
Host: www.earthlink.net
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.6) Gecko/20040113/A
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/
;q=0.8,image/png
  
```

No.	Time	Source	Destination	Protocol	Length	Info
29	4.165964	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [PSH, ACK] Seq=12421 Ack=480 Win=6432 Len=1380 [TCP PDU reassembled in 38]
30	4.216052	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=15201 Win=9660 Len=0
31	4.226976	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=13801 Ack=480 Win=6432 Len=1380 [TCP PDU reassembled in 38]
32	4.356264	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=15181 Ack=480 Win=6432 Len=1380 [TCP PDU reassembled in 38]
33	4.356264	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=16561 Win=9660 Len=0
34	4.496465	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK] Seq=16561 Ack=480 Win=6432 Len=1380 [TCP PDU reassembled in 38]
35	4.496465	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=17941 Win=9660 Len=0
36	4.776868	216.239.59.99	145.254.160.237	TCP	1484	[TCP Spurious Retransmission] 80 → 3371 [PSH, ACK] Seq=1 Ack=722 Win=31460 Len=1430
37	4.776868	145.254.160.237	216.239.59.99	TCP	54	[TCP Dup ACK 28#1] 3371 → 80 [ACK] Seq=722 Ack=1591 Win=8760 Len=0
38	4.846969	65.208.228.223	145.254.160.237	HTTP/XL	478	HTTP/1.1 200 OK
39	5.017214	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=18365 Win=9236 Len=0
40	17.905747	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [FIN, ACK] Seq=18365 Ack=480 Win=6432 Len=0
41	17.905747	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=18366 Win=9236 Len=0
42	30.063228	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [FIN, ACK] Seq=480 Ack=18366 Win=9236 Len=0
43	30.393784	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [ACK] Seq=18366 Ack=481 Win=6432 Len=0

▶ Frame 30: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
 ▶ Ethernet II, Src: Xerox_00:00:00 (00:00:01:00:00:00), Dst: fe:ff:20:00:01:00 (fe:ff:20:00:01:00:00:00)
 ▶ Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.208.228.223
 ▶ Transmission Control Protocol, Src Port: 3372, Dst Port: 80, Seq: 480, Ack: 13801, Len: 1380

Figure 6: Packet list showing packet 36 (TCP Spurious Retransmission) and packet 37 (TCP Dup ACK)

No.	Time	Source	Destination	Protocol	Length	Info
36	4.776868	216.239.59.99	145.254.160.237	TCP	1484	[TCP Spurious Retransmission] 80 → 3371 [PSH, ACK] Seq=1 Ack=722 Win=31460 Len=1430

▶ Frame 36: Packet, 1484 bytes on wire (11872 bits), 1484 bytes captured (11872 bits)
 ▶ Ethernet II, Src: fe:ff:20:00:01:00 (fe:ff:20:00:01:00:00), Dst: Xerox_00:00:00:00 (00:00:01:00:00:00)
 ▶ Internet Protocol Version 4, Src: 216.239.59.99, Dst: 145.254.160.237
 ▶ Transmission Control Protocol, Src Port: 80, Dst Port: 3371, Seq: 1, Ack: 722, Len: 1430

This frame is a (suspected) retransmission: Label

Figure 7: Filter tcp.analysis.retransmission: only packet 36 is shown (1 of 43 packets)